

## Cell 2 — Load Dataset (Notebook cell index: 4)

This cell reads training and test CSVs and renames columns to `text` and `label`.

---

### Granular line-by-line breakdown

#### Path constants

```
TRAIN_PATH = 'Training_data_9.csv'
TEST_PATH = 'Test_data.csv'
```

- `TRAIN_PATH` — Variable holding the **filename** (not used later in this exact notebook because absolute paths are hard-coded below). **Why:** Documents expected file names for readers or future refactors.
- `TEST_PATH` — Same for test file name.

**If you removed these lines:** No runtime change unless other code referenced them—here they are informational.

---

#### Read CSV files

```
train_df = pd.read_csv("I:/CSE440-Project/Training_data_9.csv")
```

- `pd` — Pandas alias from imports cell.
- `read_csv(path)` — Reads a comma-separated file into a **DataFrame** (rows = samples, columns = CSV headers).
- `"I:/CSE440-Project/..."` — **String literal:** absolute path on the author's machine. **You must change this** to where your CSV lives.
- `train_df` — Variable name holding the training DataFrame.

```
test_df = pd.read_csv("I:/CSE440-Project/Test_data.csv")
```

- Same pattern for the held-out test set `test_df`.

**If path wrong:** `FileNotFoundError`. **If CSV malformed:** pandas may error or parse incorrectly.

---

#### Rename columns

```
train_df = train_df.rename(columns={'News Headline': 'text', 'News Topic': 'label'})
```

- `train_df.rename(...)` — Returns a **new** DataFrame with columns renamed (assignment replaces `train_df` variable).
- `columns={ old: new, ... }` — Dictionary: **key** = existing header name in CSV, **value** = desired short name.
  - `'News Headline' → 'text'` — Headline string becomes column `text`.
  - `'News Topic' → 'label'` — Topic/class becomes column `label`.

```
test_df = test_df.rename(columns={'News Headline': 'text', 'News Topic': 'label'})
```

- Same mapping for test so **both** tables share schema.

**If omitted:** Code like `train_df['text']` later raises **KeyError** because columns keep

original names.

---

## Diagnostics

```
print('Train shape:', train_df.shape)
```

- `print` — Writes to notebook output.
- `'Train shape: '` — Literal label string (first argument).
- `train_df.shape` — **Attribute:** tuple (num\_rows, num\_columns).

```
print('Test shape:', test_df.shape)
```

- Same for test DataFrame.

```
print('\nLabel distribution in train:')
```

- `'\n'` — Newline before “Label distribution...” for readability.

```
print(train_df['label'].value_counts())
```

- `train_df['label']` — Series = one label per row.
- `.value_counts()` — Counts rows per distinct label (sorted by default descending).

```
train_df.head()
```

- `.head()` — Method with default **n=5**: returns first 5 rows DataFrame.
  - **Last expression in cell** — In Jupyter, displays HTML table of those rows.
- 

## Mechanism of the whole cell

Load two CSVs → rename four headers to two canonical names → print shapes and label counts → show sample rows.

---

## Role and importance in the notebook

Defines `train_df` and `test_df` with `text` and `label` for every downstream step.